EXPERIMENT 12
IMPLEMENTATION OF RSA ALGORITHM

AIM

To implement the RSA algorithm for encryption and decryption using public and private keys.

ALGORITHM

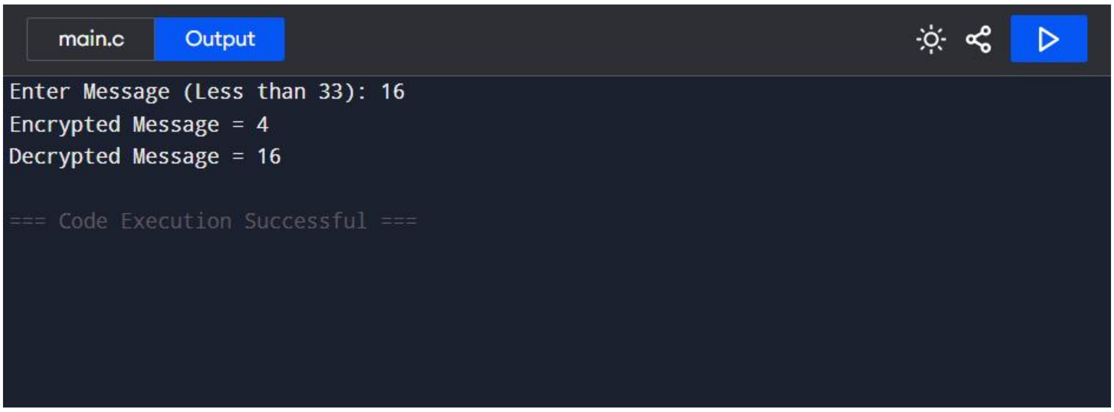
1. Choose two prime numbers p and q.
2. Calculate $n = p \times q$.
3. Choose the public key e and private key d.
4. Read the plaintext message.
5. Encrypt the message using the public key.
6. Decrypt the ciphertext using the private key.

CODE

```
#include <stdio.h>

long long power(long long a,long long b,long long m)
{
   long long r=1;
   while(b>0)
   {
      r=(r*a)%m;
      b--;
   }
   return r;
}

int main()
{
   int p=3,q=11,e=3,d=7,n,msg;
   long long c,m;

   n=p*q;

   printf("Enter Message (Less than %d): ",n);
   scanf("%d",&msg);

   c=power(msg,e,n);
   printf("Encrypted Message = %lld\n",c);
```

```
    m=power(c,d,n);
    printf("Decrypted Message = %lld",m);

    return 0;
}
```

OUTPUT



RESULT

The message was successfully encrypted using the public key and decrypted using the private key using the RSA algorithm.